EXERCISE SHEET 3

GALOIS THEORY

3.1. Is the following finite extension $L \supseteq K$ normal?

$K = \mathbb{Z}/3\mathbb{Z}$, $L = (\mathbb{Z}/3\mathbb{Z})(i)$ (here $i^2 = -1$ and L is a field with 9 elements).

3.2. Prove that every extension of degree 2 is normal.

3.3. Assuming that all polynomials split over $\mathbb{C}$, prove that the set $A(\mathbb{C}, \mathbb{Q})$ of all algebraic numbers over $\mathbb{Q}$ is a field that has no finite extensions. Prove that every polynomial over $A(\mathbb{C}, \mathbb{Q})$ splits over $A(\mathbb{C}, \mathbb{Q})$. [You may use the result of the next exercise.]

3.4. Let $L : K$ be an arbitrary field extension. Prove that the set $A(L, K)$ of all elements of L that are algebraic over K is a subfield of L. [*Hint:* Use the theorem on finite extensions.]

3.5. Prove the basic lemma on K-monomorphisms. In other words, suppose that $K \subseteq L \subseteq M$ are fields and L is a splitting field for $f \in K[t]$. Show that any K-automorphism φ of M permutes the roots of f and leaves L invariant: $f(a) = 0 \Rightarrow f(\varphi(a)) = 0$ and $\varphi(L) = L$.

3.6. Let L be a splitting field of a nonconstant polynomial $f \in K[t]$. Then $\mathrm{Gal}(L/K)$ is finite. In particular, if $a_1, \ldots, a_n$ are the distinct roots of f in L, then $\mathrm{Gal}(L/K)$ is isomorphic to a subgroup of the group of permutations of $\{a_1, \ldots, a_n\}$, so its order divides $n!$.

3.7. Let ω be a non-real fifth root of 1 (e. g., $\omega = e^{2\pi i/5}$). Prove that G, the Galois group of $\mathbb{Q}(\omega) : \mathbb{Q}$, is cyclic of order 4. [*Hints:*

(a) Show that ω, ω^2, ω^3, ω^4 are the roots of the irreducible polynomial $t^4 + t^3 + t^2 + t + 1$;

(b) if $g \in G$, then $g(\omega) = \omega^i$ for $i = 1, \ldots, 4$ and g is determined by $g(\omega)$;

(c) for each $j = 1, \ldots, 4$ there is an element $g_j \in G$ with $g_j(\omega) = \omega^j$ (by which theorem?).]